THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 13: **Our Home Earth, a Unique Life-Sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** distance from the Sun, atmosphere and water	Several conditions together
2	**(b)** escape into space	Gravity holds air
3	**(d)** high-energy particles from space	Protective shield
4	**(c)** water	Oceans, rivers, ice
5	**(a)** continuity of life	Species survive
6	**(d)** the first cell formed after fertilisation	Develops into embryo
7	**(c)** Planaria	Regeneration studies
8	**(b)** inside the female's body	Internal development
9	**(c)** methane	Trap heat
10	**(c)** makes ecosystems stronger	Diverse ecosystems are balanced

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Region around a star where water stays liquid (1). Also called Goldilocks zone (1).	1+1
12	Right distance from the Sun (1). Atmosphere with oxygen and ozone (1).	1+1
13	Holds the atmosphere (1). Not too strong to crush creatures (1).	1+1
14	Absorbs harmful ultraviolet rays (1). Protects life on Earth's surface (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Water stays liquid (1½). Temperature is neither too hot nor too cold (1½).	1½+1½
16	Habitable zone position (1). Gravity holding atmosphere (1). Ozone and magnetic field protect (1).	1+1+1
17	Gases trap heat (1). Keeps Earth warm (1). Without it Earth would be too cold (1).	1+1+1
	OR: Venus has dense carbon dioxide atmosphere (1). Strong greenhouse effect makes it very hot (1). Earth's effect keeps a moderate temperature (1).	1+1+1
18	Water evaporates from hydrosphere (1). Clouds form in atmosphere and rain falls on land (1). Plants use water, supporting biosphere (1).	1+1+1
	OR: Plants grow in it (1). Provides minerals and nutrients (1). Home for microorganisms and animals (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Position in habitable zone (1). Nearly circular orbit (1). Right gravity (1). Ozone layer (1). Magnetic field (1).	1+1+1+1+1
	OR: Gases trap heat (1). Keeps Earth warm (1). Burning fossil fuels raises CO_2 (1). Methane from other sources (1). Global temperatures rise (1).	1+1+1+1+1
20	Atmosphere (1). Hydrosphere (1). Geosphere (1). Biosphere (1). Interactions with example (1).	1+1+1+1+1
	OR: Single parent (1). Identical offspring (1). Stem cutting (1). Regeneration (1). Examples of animals (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 13: **Our Home Earth, a Unique Life-Sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** habitable zone	Goldilocks zone
2	**(a)** crushed by their own weight	Too much pull
3	**(b)** keeps Earth warm by trapping heat	Natural, essential
4	**(d)** solid part of the Earth	Crust, rocks, minerals
5	**(b)** a single parent produces offspring identical to itself	Exact copies
6	**(c)** pollination	By wind, insects, birds
7	**(c)** in water	External fertilisation
8	**(d)** eaten by birds and dropped	Dispersal
9	**(c)** disturb climate patterns	Serious consequences
10	**(b)** geosphere	Solid Earth

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Shields Earth from high-energy particles (1). Protects the atmosphere and living things (1).	1+1
12	Atmosphere – air (1). Hydrosphere – water, geosphere – solid Earth, biosphere – life (any one) (1).	1+1
13	Atmosphere traps some of the Sun's heat (1). Keeps Earth warm enough for life (1).	1+1
14	Carbon dioxide (1). Methane (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Part of plant such as stem or leaf grows into new plant (1½). Sugarcane, rose, potato (1½).	1½+1½
16	Regrowth of a body from a small piece (1½). Planaria (1½).	1½+1½
17	Pollen transferred by wind, insects or birds (1½). Gametes fuse to form zygote then seed (1½).	1½+1½
	OR: Seeds are carried to other places (1). Birds eating fruits drop seeds (1). Banyan seeds sprout in cracks (1).	1+1+1
18	Fish: fertilisation in water (1½). Mammals: fertilisation and development inside the body (1½).	1½+1½
	OR: Fertilisation inside the female (1). Zygote enclosed in egg and laid (1). Embryo develops during hatching (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pollination (1). Gametes (1). Fertilisation (1). Zygote to seed (1). Seed dispersal and germination (1).	1+1+1+1+1
	OR: Fish and frogs fertilise in water (1½). Embryo develops in water (1). Birds fertilise inside, egg laid (1½). Mammals develop inside (1).	1½+1+1½+1
20	Climate change (1). Pollution (1). Biodiversity loss (1). Individual actions (1). Community and global action (1).	1+1+1+1+1
	OR: Oxygen for breathing (1). Ozone absorbs UV (1). Greenhouse effect warms (1). Protects from space particles (1). Balanced composition (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.